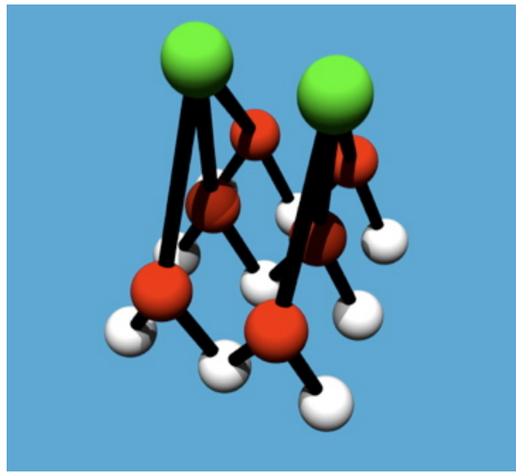
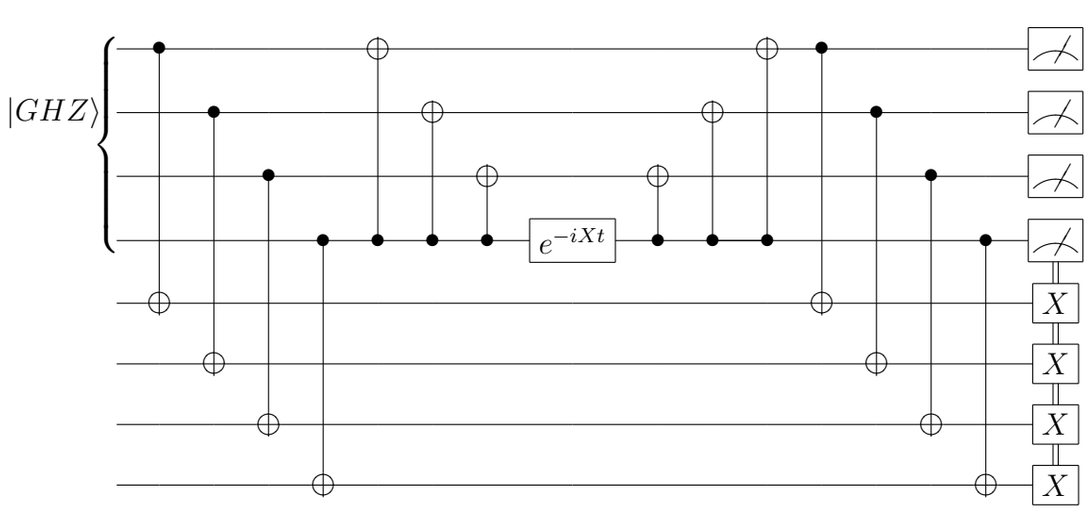
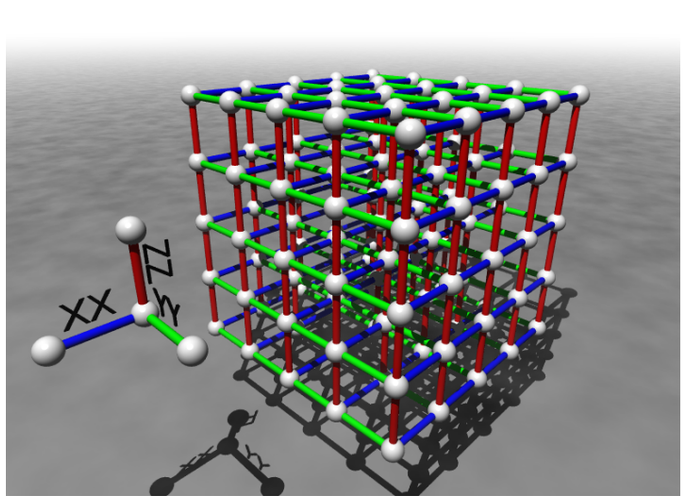
Robust Quantum Gadgets

Dave Bacon
Department of Computer Science & Engineering
Department of Physics
University of Washington



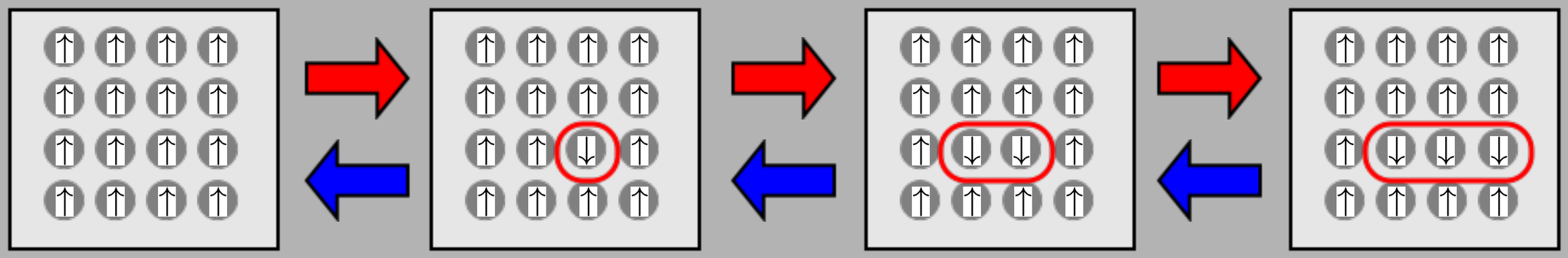




"Jump starting a new path to robust quantum devices"

Self-Correcting Computers

Why does you hard drive robustly store information?



Spin flip errors cost energy proportional to the perimeter of the flipped region

At low enough temperature, "cooling" ⬅ wins over "heating" ➡ and the information in the majority vote of the spins is preserved in spite of the noise

Hard drives use **physics** to protect classical information by **naturally implementing error correction**

Self-Correcting Quantum Computers

Use physics to protect quantum information by **naturally implementing quantum error correction**



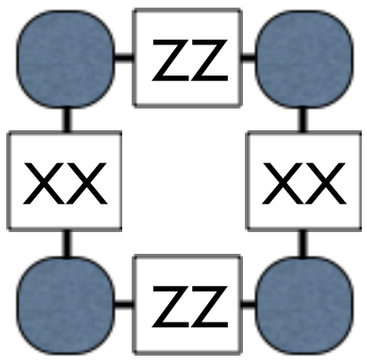
How: Build many-body interacting quantum systems based around quantum error correcting codes with energy barriers to provide self-correction (this project: ways to do this)

Why: Replaces active quantum error correction
Lowers device complexity
Less resources (no measurement, no timing)
Avoids scalability issue

Impact: Totally new method to achieve fault-tolerant quantum computing. End-around on traditional QC roadmap.

Subsystem Stabilizer Hamiltonians

few examples of energy stabilized quantum systems with realistic two qubit interactions are all subsystem code constructions



$$H = X_1(I + S_1) + Z_1(I + S_2)$$

subsystem code

$$\begin{aligned} S_1 &= X \otimes X \otimes X \otimes X \\ S_2 &= Z \otimes Z \otimes Z \otimes Z \end{aligned}$$ stabilizer

$$\begin{aligned} X_1 &= X \otimes X \otimes I \otimes I \\ Z_1 &= I \otimes Z \otimes Z \otimes I \end{aligned}$$ gauge

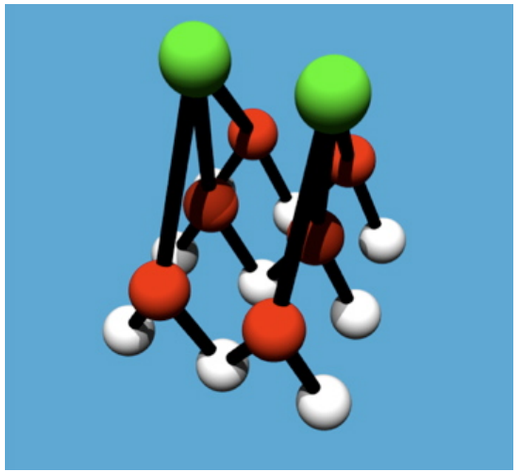
$$\begin{aligned} X_2 &= I \otimes X \otimes X \otimes I \\ Z_2 &= Z \otimes Z \otimes I \otimes I \end{aligned}$$ logical

TASK: systematic analytic and numeric study of subsystem code Hamiltonians generated by 2-local interactions

POSITIVE RESULT: few qubit systems suitable for implementing basic self-correction in ten to twenty qubit systems.

SYNERGY: PI identified first nontrivial subsystem codes and developed original notion of "supercoherent" qubits

Robust Perturbation Theory Gadgets



$$H = H_0 + \lambda V$$
$$H_0 = \Delta(Z_a Z_b + Z_b Z_c + Z_c Z_a)$$
$$V = X_1 X_a + X_2 X_b + X_3 X_b$$

$$H_{eff} = c\frac{\lambda^3}{\Delta^2} X_1 X_2 X_3 X_a X_b X_c + O\left(\frac{\lambda^4}{\Delta^3}\right)$$

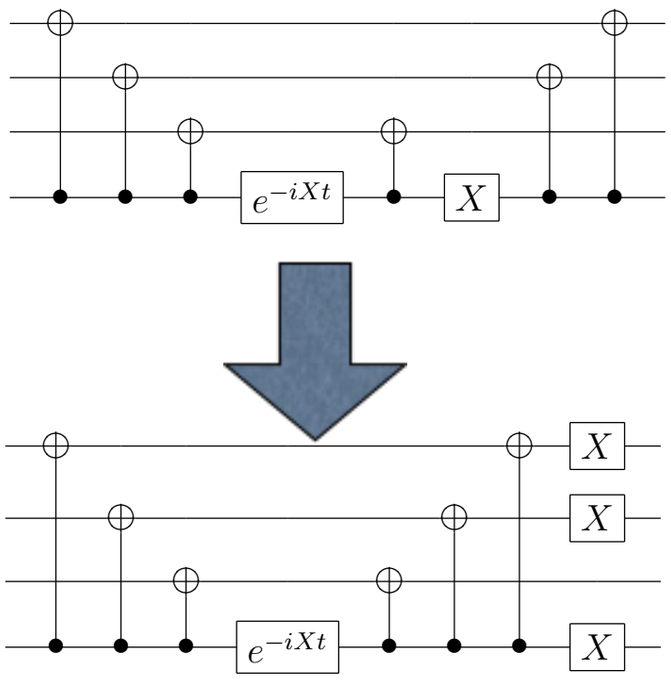
TASK: development of new perturbation theory gadgets for implementing stabilizer Hamiltonians with greater than two qubit interactions and making them fault-tolerant

POSITIVE RESULT: methods for engineering strongly interacting quantum systems with quantum error correcting order parameters

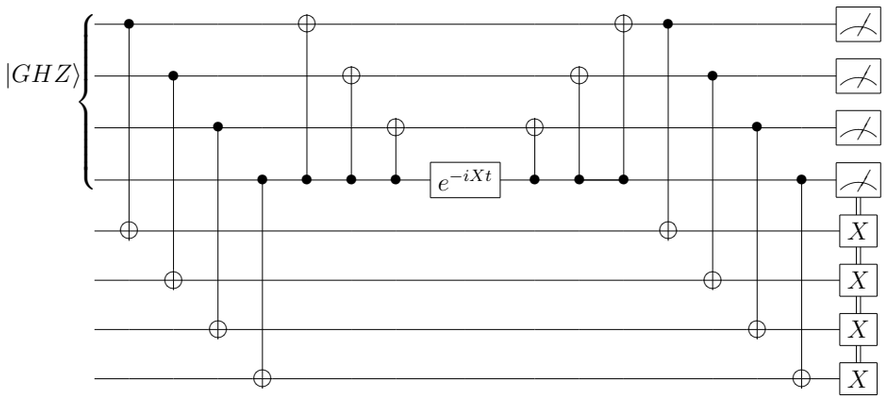
SYNERGY: PI and physics graduate student have implemented matrix product state-like methods for simulation of hundred qubit quantum systems.

Bang-Bang Simulation of Stabilizer Hamiltonians

not fault-tolerant:



fault-tolerant:



TASK: develop fault-tolerant bang-bang simulation techniques for trotter simulation of stabilizer Hamiltonians

POSITIVE RESULT: render simulation techniques suitable for self-correcting quantum computation

SYNERGY: PI has background in fault-tolerant quantum computation as well as in quantum simulation techniques